Gnana Prakash

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Summary

Graduate Electrical and Computer Engineering student specializing in embedded systems and biomedical signal processing. Experienced in designing real-time sensing systems, neural interfaces, and edge AI pipelines using MATLAB, Python, and microcontrollers. Strong focus on EEG-based monitoring, physiological sensing, and safety-critical system design. Seeking roles or research opportunities in embedded systems, neurotechnology, and real-time signal processing.

Projects

AlphaSync Bio (EEG Fatigue Monitoring)
- Designed EEG-based pipeline for real-time cognitive fatigue estimation
- Implemented signal filtering and feature extraction using MATLAB
- Targeted safety-critical applications in neuroergonomics

Acoustic Heart Rate Detection (ESP32 + INMP441)
- Developed real-time heartbeat detection using I2S microphone input
- Implemented embedded classification achieving >90% accuracy on real-time signals
- Designed on-device inference pipeline with live feedback

Neural Switching System (16-channel)
- Designed high-voltage (100–120V) switching architecture for neural interfaces
- Implemented MCU-controlled electrode routing with hardware-level protection logic
- Enabled safe simultaneous stimulation and recording across multi-channel system

Wearable Respiratory Monitor
- Developed sensor fusion pipeline using MPU6050 and AHT10
- Extracted breathing patterns using filtered motion signals
- Designed LoRa-enabled wearable monitoring system

Work Experience

University of Alabama at Birmingham **Jan 2026 – Present**
Graduate Research Assistant — ANRY Lab
- Designed a 16-channel neural switching system routing electrodes between Ripple Grapevine EEG and STG4002-opto.
- Implemented high-voltage (100–120 V) switching via relays/PhotoMOS with MCU + 74HC595 sequencing.
- Developed TTL-triggered stimulation control, timing logic, and Multisim-based protection circuits.

UAB Campus Recreation Center **May 2025 - Aug 2025**
Camp Counselor
- Supervised group activities and ensured safety in a structured environment

University of Alabama at Birmingham **Jan 2025 - Apr 2025**
Teaching Assistant
- Assisted in delivering lectures and grading assignments for embedded systems courses
- Mentored students on hardware debugging and system design

University of Alabama at Birmingham **Aug 2024 - Dec 2024**
Graduate Research Assistant
- Developed MATLAB-based EEG signal processing pipeline improving signal clarity and feature extraction accuracy for neural analysis.
- Developed computational models for artifact removal in brain-machine interface research, leading to more reliable data interpretation
- Conducted advanced signal processing techniques to improve EEG and neural stimulation data analysis, resulting in more precise data insights

University of Alabama at Birmingham **Feb 2024 - Aug 2024**
Graduate Research Assistant
- Developed a data collection model using Power Automate to streamline automation processes in industrial applications, improving data accuracy and reducing processing time
- Preprocessed collected data using Python and performed advanced data visualization with Matplotlib and Seaborn, leading to more informed decision-making in research projects
- Conducted research on real-time system control optimization to enhance industrial efficiency, resulting in improved system performance and reduced operational costs
- Designed and implemented a multitasking system using FreeRTOS on Arduino, managing hardware synchronization for reliable embedded system operations, which increased system reliability and performance

University Recreation Center **May 2024 - May 2025**
Operations Attendant
- Collaborated with cross-functional teams to troubleshoot and resolve equipment issues, ensuring operational efficiency while adhering to safety standards.
- Contributed to improving workflow processes, leading to a 10% increase in overall productivity.

Education

University of Alabama at Birmingham **Jan 2024 - Dec 2025**
Masters, Electrical and computer engineering- 3.5 ***GPA***

RGUKT **Jan 2019 - Jan 2023**
UG, ECE - 7.8 **GPA** *IIIT Srikakulam,INDIA*

Skills

Programming: C++, Python, MATLAB, Verilog
Embedded Systems: FreeRTOS, ARM, Arduino, MCU design
Signal Processing: EEG analysis, filtering, feature extraction
Hardware: Circuit design, power electronics, Multisim
Tools: AutoCAD, MATLAB/Simulink